

Power Calculation for Case–Cohort Studies with Nonrare Events

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SUMMARY. Case–cohort design has been advocated in many epidemiologic studies when studying rare diseases or events. In this design, with a rare event, all the events are selected for risk-factor assessment. When the event is not rare, it is desirable to consider a generalized case–cohort design, where only a fraction of events are sampled. We provide a valid test statistic to compare hazards functions between two samples for this generalized design and give a method for calculating power. Our result generalizes the result in Cai and Zeng (2004, *Biometrics* **60**, 1015–1024), and it shows numerically that efficiency loss due to sampling only part of the events is very low under nonrare-events situation.

KEY WORDS: Case–cohort design; Generalized case–cohort design; Nonrare events; Power calculation; Rare events.

1. Introduction

The case–cohort design has been extensively used in epidemiologic studies when the study endpoint is time to event (e.g., Schouten et al., 1993; Liao et al., 1997; Savitz et al., 2000; Folsom et al., 2001). In such a design, all the failures and a random sample of such cohort are selected to assess the risk-factor information. Since its proposal, the case–cohort design has been used in many studies. Many methods have been developed to study such a design, for instance, Prentice (1986), Self and Prentice (1988), Lin and Ying (1993), and Borgan, Goldstein, and Langholz (1995) among others.

The design can be almost as efficient as the full cohort design in which all the subjects in the cohort are selected for risk-factor assessment, even when the selected subcohort consists of a small portion of the whole cohort. Case–cohort design is advocated especially when failure rate is low and measuring covariates is expensive. A recent paper by Cai and Zeng (2004) provides a simple formula for sample size/power calculation under rare failure situation. However, in many studies, the failure rate may not be low and the number of failures could be large. Under such circumstances, it is desirable to assemble covariates information for only a subset of the failures instead of all the failures due to financial constraints. In this situation, it is uncertain whether the sample size/power formula in Cai and Zeng (2004) is still applicable. Thus, this motivates us to seek simple formulae applicable to general situations.

We consider the following generalized case–cohort design: in the first step, we randomly sample without replacement a subcohort from the full cohort; in the second step, another sample is randomly taken without replacement from the remaining failures. The sampling fraction in this step is referred

to as the failure sampling fraction. The sample obtained from these two steps constitutes the generalized case–cohort sample, and the covariate information is only assembled on this sample. Obviously, when the failure sampling fraction is one, the design becomes the classical case–cohort design. When the failure sampling fraction is small, we only sample a fraction of failures—so this is useful when failure rate is not low.

For the generalized case–cohort design, we will propose a general log-rank test statistic and derive its asymptotic property. Our construction of the test statistic is essentially to approximate the risk set and the event process of the full cohort using the sampled data. Based on the proposed test statistic, we will also provide a way of computing power, which can be applied to both rare and nonrare failures and can adapt to complicated censoring situations.

The article is organized as follows. In Section 2, we will propose a general test for the generalized case–cohort design and provide its asymptotic properties. In Section 3, we will derive the power calculation for the proposed test. In Section 4, numerical studies will be conducted to evaluate the small-sample performance of the proposed test and compare with the result of the full cohort. Some discussion and further consideration will be given in Section 5.

2. Test Statistic for Generalized Case–Cohort Design

In this section, we provide a test statistic for a generalized case–cohort design, where a subcohort is sampled and then a fraction is sampled from the remaining cases. The main difference of the generalized case–cohort design from the usual case–cohort design is that not all the remaining cases are sampled for exposure assessment.

We first define some notations. Let n denote the number of subjects in the entire cohort. Let X_i be the dichotomous variable indicating the exposure status, Z_i be the observed event time, and Δ_i be the right-censoring indicator. Let p denote the subcohort sampling probability and ξ_i indicate the selection of subject i . Similarly, let q denote the failure sampling probability and η_i indicate the selection of subject i . With this notation, the subcohort size has mean np and the number of sampled additional failures has conditional mean $\tilde{n}q$, where $\tilde{n}$ denotes the total number of failures not in the subcohort, i.e., $\xi_i = 0$ and $\Delta_i = 1$. We note that in the traditional case-cohort design, $q = 1$. The observed data can be expressed as

$$(X_i(\xi_i + (1 - \xi_i)\Delta_i\eta_i), Z_i, \Delta_i), \quad i = 1, \dots, n,$$

where $(\xi_i + (1 - \xi_i)\Delta_i\eta_i)$ is the indicator for whether X_i is observable.

We assume that X is a dichotomous factor. Let $X = 1$ denote group 1 and $X = 2$ denote group 2. To compare the two groups, we construct a similar test statistic to the usual log-rank test as follows:

$$W_n = \int_0^\tau \frac{\omega(t)\tilde{Y}_1(t)\tilde{Y}_2(t)}{\tilde{Y}_1(t) + \tilde{Y}_2(t)} \left\{ \frac{d\tilde{N}_1(t)}{\tilde{Y}_1(t)} - \frac{d\tilde{N}_2(t)}{\tilde{Y}_2(t)} \right\},$$

where τ is the study duration, $\omega(t)$ is a weight function, and for $j = 1, 2$,

$$\begin{aligned} \tilde{Y}_j(t) &= \sum_{i=1}^n I(X_i = j, Z_i \geq t) \\ &\quad \times \left[\xi_i \Delta_i + \frac{\xi_i(1 - \Delta_i)}{p} + \frac{\Delta_i(1 - \xi_i)\eta_i}{q} \right] \\ &= \sum_{i=1}^n I(X_i = j, Z_i \geq t) \\ &\quad \times \left[\frac{\Delta_i(1 - \xi_i)\eta_i}{q} + \left(\Delta_i + \frac{1 - \Delta_i}{p} \right) \xi_i \right], \\ \tilde{N}_j(t) &= \sum_{i=1}^n \Delta_i I(X_i = j, Z_i \leq t) \left\{ \xi_i + \frac{(1 - \xi_i)\eta_i}{q} \right\}. \end{aligned}$$

Note that in calculating both $\tilde{Y}_j(t)$ and $\tilde{N}_j(t)$, there is no contribution from subjects not sampled because ξ_i and η_i take zero values for these subjects. Indeed, $\tilde{Y}_j(t)$ can be treated as the risk set of the full cohort in group j approximated by the sampled subjects; that is, each sampled noncase in the subcohort represents $1/p$ subjects in the noncases of the full cohort while each sampled additional case outside the subcohort represents $1/q$ subjects in the cases that are not sampled in the subcohort. Similarly, $\tilde{N}_j(t)$ can be considered as the counting process of the full cohort in group j approximated by the sampled cases. Moreover, when $q = 1$ in the traditional case-cohort design, $\tilde{Y}_j(t) = \sum_{i=1}^n I(X_i = j, Z_i \geq t)[\Delta_i + \xi_i(1 - \Delta_i)/p]$ and $\tilde{N}_j(t) = \sum_{i=1}^n \Delta_i I(X_i = j, Z_i \leq t)$. Thus, W_n reduces to the statistic V_n , which was studied in Cai and Zeng (2004).

The following theorem gives the asymptotic properties of W_n . Its proof is given in the Web Appendix.

THEOREM 1: Denote $\alpha(t) = P(X = 1, C \geq t)/P(C \geq t)$, where C denotes the right-censoring time. Under the null hypothesis that two groups have the same cumulative hazards function $\Lambda(t)$, $n^{-1/2}W_n$ has an asymptotic normal distribution with mean zero and variance equal to

$$\begin{aligned} \sigma_W^2 &= \sigma^2 + \frac{1-p}{p} \text{Var}(\nu(Z, X)(1 - \Delta)) \\ &\quad + \frac{P(\Delta = 1)(1-p)(1-q)}{q} \text{Var}(\mu(Z, X) | \Delta = 1, \xi = 0), \end{aligned}$$

where σ^2 is the asymptotic variance of the weighted log-rank statistic of the full cohort design and

$$\begin{aligned} \nu(Z, X) &= - \int_0^Z \omega(t)(1 - \alpha(t)) d\Lambda(t) I(X = 1) \\ &\quad + \int_0^Z \omega(t)\alpha(t) d\Lambda(t) I(X = 2), \end{aligned}$$

and

$$\begin{aligned} \mu(Z, X) &= \nu(Z, X) + \omega(Z)(1 - \alpha(Z))I(X = 1) \\ &\quad - \omega(Z)\alpha(Z)I(X = 2). \end{aligned}$$

We can consistently estimate σ_W^2 using the observed data. First, σ^2 , the variance of the log-rank statistic for the full cohort, can be estimated by

$$\hat{\sigma}^2 = \frac{1}{n} \left\{ \int_0^\tau \frac{\tilde{Y}_2(t)^2 d\tilde{N}_1(t)}{(\tilde{Y}_1(t) + \tilde{Y}_2(t))^2} + \int_0^\tau \frac{\tilde{Y}_1(t)^2 d\tilde{N}_2(t)}{(\tilde{Y}_1(t) + \tilde{Y}_2(t))^2} \right\}.$$

Additionally, if we define

$$\begin{aligned} \hat{\mu}(Z, X) &= \int_0^Z \frac{\omega(t)\{\tilde{Y}_1(t)I(X=2) - \tilde{Y}_2(t)I(X=1)\}\{d\tilde{N}_1(t) + d\tilde{N}_2(t)\}}{(\tilde{Y}_1(t) + \tilde{Y}_2(t))^2} \\ &\quad - \frac{\omega(Z)\{\tilde{Y}_1(Z)I(X=2) - \tilde{Y}_2(Z)I(X=1)\}}{\tilde{Y}_1(Z) + \tilde{Y}_2(Z)}, \end{aligned}$$

and

$$\hat{\nu}(Z, X) = \int_0^Z \frac{\omega(t)\{\tilde{Y}_1(t)I(X=2) - \tilde{Y}_2(t)I(X=1)\}\{d\tilde{N}_1(t) + d\tilde{N}_2(t)\}}{(\tilde{Y}_1(t) + \tilde{Y}_2(t))^2},$$

then it is easy to see that $\hat{\mu}(Z, X)$ and $\hat{\nu}(Z, X)$ uniformly converge to $\mu(Z, X)$ and $\nu(Z, X)$, respectively. Therefore, we can consistently estimate the asymptotic variance of $n^{-1/2}W_n$ as

$$\begin{aligned} \hat{\sigma}_W^2 &= \hat{\sigma}^2 + \frac{1-p}{p} \left[\frac{1}{np} \sum_{i=1}^n \hat{\nu}(Z_i, X_i)^2 (1 - \Delta_i) \xi_i \right. \\ &\quad \left. - \left\{ \frac{1}{np} \sum_{k=1}^n \hat{\nu}(Z_k, X_k) (1 - \Delta_k) \xi_k \right\}^2 \right] \\ &\quad + \frac{\tilde{n}(1-q)}{nq} \left[\frac{1}{\tilde{n}q} \sum_{i=1}^n \hat{\mu}(Z_i, X_i)^2 \Delta_i (1 - \xi_i) \eta_i \right. \\ &\quad \left. - \left\{ \frac{1}{\tilde{n}q} \sum_{k=1}^n \hat{\mu}(Z_k, X_k) \Delta_k (1 - \xi_k) \eta_k \right\}^2 \right], \end{aligned}$$

where $\tilde{n}$ denotes the number of events not sampled in the subcohort. The latter is because

$$\begin{aligned} \frac{1}{\tilde{n}q} \sum_{i=1}^n \hat{\mu}(Z_i, X_i)^2 \Delta_i (1 - \xi_i) \eta_i &\rightarrow \frac{1}{q} E[\mu(Z, X)^2 \eta | \Delta = 1, \xi = 0] \\ &= E[\mu(Z, X)^2 | \Delta = 1, \xi = 0] \end{aligned}$$

and

$$\begin{aligned} \frac{1}{\tilde{n}q} \sum_{i=1}^n \hat{\mu}(Z_i, X_i) \Delta_i (1 - \xi_i) \eta_i &\rightarrow \frac{1}{q} E[\mu(Z, X) \eta | \Delta = 1, \xi = 0] \\ &= E[\mu(Z, X) | \Delta = 1, \xi = 0], \end{aligned}$$

and similarly,

$$\frac{1}{np} \sum_{i=1}^n \hat{\nu}(Z_i, X_i)^2 (1 - \Delta_i) \xi_i \rightarrow E[\nu(Z, X)^2 (1 - \Delta)]$$

and

$$\frac{1}{np} \sum_{i=1}^n \hat{\nu}(Z_i, X_i) (1 - \Delta_i) \xi_i \rightarrow E[\nu(Z, X) (1 - \Delta)].$$

Hence, for the one-sided test in which group 1 has a larger hazard risk, we reject the null hypothesis when $n^{-1/2} W_n / \hat{\sigma}_W > \Phi^{-1}(1 - \alpha)$ at the significance level α .

3. Power Calculation for Generalized Case-Cohort Design

In this section, we propose a method for calculating the power for the generalized design. To make the calculation simple, we assume $\omega(t) = 1$ in the test statistic W_n . Furthermore, we assume $P(X = 1) = \gamma$.

The power is calculated under a local alternative hypothesis where the cumulative hazard function in group 1 is $\Lambda_1(t) = \Lambda(t)$ and the cumulative hazard function in group 2 is $\Lambda_2(t) = \Lambda(t) \exp\{-\theta\}$, where θ is a positive constant of order $n^{-1/2}$. From the proof of Theorem 1, the asymptotic noncentrality parameter of $n^{-1/2} W_n$ is approximated by

$$\begin{aligned} n^{-1/2} \int_0^\tau \alpha(t) \tilde{Y}_2(t) \{d\Lambda_1(t) - d\Lambda_2(t)\} \\ = \sqrt{n}(1 - \gamma)\theta \int_0^\tau \alpha(t) P(C \geq t | X = 2) e^{-\Lambda(t)} d\Lambda(t) + o_p(1). \end{aligned}$$

Therefore, it remains to simplify the expression of σ_W^2 .

To simplify the calculation, we assume $P(C > t | X = 2) = \beta P(C > t | X = 1)$ and denote $P(C > t | X = 1) = S_c(t)$, i.e., the subjects in group 2 is β -times as likely to be censored as in group 1. After tedious derivation as given in the Web Appendix, σ_W^2 can be simplified as

$$\begin{aligned} \sigma_W^2 &= \frac{\beta\gamma(1 - \gamma)}{\gamma + (1 - \gamma)\beta} \left\{ 1 + \frac{(1 - p)(1 - q)}{q} \right\} \int_0^\tau S_c(t) e^{-\Lambda(t)} d\Lambda(t) \\ &\quad + 2 \frac{\beta\gamma(1 - \gamma)}{\gamma + (1 - \gamma)\beta} \left\{ \frac{1 - p}{p} - \frac{(1 - p)(1 - q)}{q} \right\} \\ &\quad \times \int_0^\tau S_c(t) \Lambda(t) e^{-\Lambda(t)} d\Lambda(t) + \frac{\beta\gamma(1 - \gamma)}{\gamma + (1 - \gamma)\beta} \\ &\quad \times \left\{ -\frac{1 - p}{p} + \frac{(1 - p)(1 - q)}{q} \right\} \int_0^\tau S_c(t) \Lambda(t)^2 e^{-\Lambda(t)} d\Lambda(t). \end{aligned}$$

Clearly, σ_W^2 depends on the censoring distribution. Thus, if $S_c(t)$ is known, the above formula can be used to compute the power.

We wish to estimate σ_W^2 using the observed quantities. First, we note that $Y(t)/n(\gamma + (1 - \gamma)\beta) \rightarrow S_c(t) \exp\{-\Lambda(t)\}$ and $\int_0^t dN(s)/Y(s) \rightarrow \Lambda(t)$ where $Y(t) = Y_1(t) + Y_2(t)$ and $N(t) = N_1(t) + N_2(t)$. Therefore, we can approximate σ_W^2 by

$$\begin{aligned} \sigma_W^2 &\approx \frac{\beta\gamma(1 - \gamma)}{\gamma + (1 - \gamma)\beta} \left\{ 1 + \frac{(1 - p)(1 - q)}{q} \right\} \frac{1}{n(\gamma + (1 - \gamma)\beta)} \int_0^\tau dN(t) \\ &\quad + 2 \frac{\beta\gamma(1 - \gamma)}{\gamma + (1 - \gamma)\beta} \left\{ \frac{1 - p}{p} - \frac{(1 - p)(1 - q)}{q} \right\} \frac{1}{n(\gamma + (1 - \gamma)\beta)} \\ &\quad \times \int_0^\tau \int_0^t \frac{dN(s)}{Y(s)} dN(t) + \frac{\beta\gamma(1 - \gamma)}{\gamma + (1 - \gamma)\beta} \left\{ -\frac{1 - p}{p} + \frac{(1 - p)(1 - q)}{q} \right\} \\ &\quad \times \frac{1}{n(\gamma + (1 - \gamma)\beta)} \int_0^\tau \left\{ \int_0^t \frac{dN(s)}{Y(s)} \right\}^2 dN(t). \end{aligned}$$

We suppose that no two failures are observed to be tied so we order them from the smallest to the largest. For total D failures, σ_W^2 is approximated by

$$\begin{aligned} \sigma_W^2 &\approx \frac{\beta\gamma(1 - \gamma)}{(\gamma + (1 - \gamma)\beta)^2} \left\{ 1 + \frac{(1 - p)(1 - q)}{q} \right\} \frac{D}{n} \\ &\quad + 2 \frac{\beta\gamma(1 - \gamma)}{(\gamma + (1 - \gamma)\beta)^2} \left\{ \frac{1 - p}{p} - \frac{(1 - p)(1 - q)}{q} \right\} \\ &\quad \times \frac{1}{n} \sum_{k=1}^D \sum_{k' \leq k} \frac{1}{n_{k'}} + \frac{\beta\gamma(1 - \gamma)}{(\gamma + (1 - \gamma)\beta)^2} \\ &\quad \times \left\{ -\frac{1 - p}{p} + \frac{(1 - p)(1 - q)}{q} \right\} \frac{1}{n} \sum_{k=1}^D \left\{ \sum_{k' \leq k} \frac{1}{n_{k'}} \right\}^2, \end{aligned}$$

where D is the number of failures and n_k is the risk set size for k th failure in the full cohort. On the other hand, similar approximation gives that the asymptotic mean of $n^{-1/2} W_n$ is approximated by

$$\text{mean} \approx \sqrt{n} \frac{\beta\gamma(1 - \gamma)\theta}{\gamma + (1 - \gamma)\beta} p_D,$$

where $p_D = D/n$ is the observed failure rate after censoring. Consequently, the power based on the proposed test statistic is given by

$$\text{Power} = \Phi \left(Z_\alpha + \frac{\sqrt{n\beta\gamma(1 - \gamma)\theta p_D}}{\sqrt{\chi}} \right), \quad (1)$$

where

$$\begin{aligned}
\chi &= \left\{ 1 + \frac{(1-p)(1-q)}{q} \right\} p_D + \left\{ \frac{1-p}{p} - \frac{(1-p)(1-q)}{q} \right\} \\
&\quad \times \frac{1}{n} \left\{ 2 \sum_{k=1}^D \sum_{k' \leq k} \frac{1}{n_{k'}} - \sum_{k=1}^D \left\{ \sum_{k' \leq k} \frac{1}{n_{k'}} \right\}^2 \right\} \\
&= \frac{p_D}{p} - \left\{ \frac{1-p}{p} - \frac{(1-p)(1-q)}{q} \right\} \frac{1}{n} \sum_{k=1}^D \left\{ 1 - \sum_{k' \leq k} \frac{1}{n_{k'}} \right\}^2.
\end{aligned} \tag{2}$$

It is easy to see that the power increases with q .

In many practical designs, the censoring distribution, i.e., the size of risk sets $\{n_k\}$, is unknown. We propose the following lower bound and upper bound for the power calculation: Note that $n - n_c - D \leq n_k \leq n - k + 1$, where n_c is the total number of dropouts in $[0, \tau)$. Thus, we can calculate the two bounds as follows: consider $(D + 1)$ possibilities of extreme sequences that for $k = 1, \dots, D$, we set

$$\begin{aligned}
n_1 &= n, n_2 = n - 1, \dots, n_k = n - k + 1, \\
n_{k+1} &= \dots = n_D = n - n_c - D;
\end{aligned}$$

while for $k = 0$, we let all risk size be $n - n_c - D$. We then evaluate χ for each of these sequences. Define χ_L and χ_U as the lower bound and upper bound of these χ 's. Then we conclude

$$\begin{aligned}
&\Phi \left(Z_\alpha + \frac{\sqrt{n\beta\gamma(1-\gamma)\theta p_D}}{\sqrt{\chi_U}} \right) \\
&\leq \text{Power} \leq \Phi \left(Z_\alpha + \frac{\sqrt{n\beta\gamma(1-\gamma)\theta p_D}}{\sqrt{\chi_L}} \right).
\end{aligned}$$

This gives the range of power based on the proposed test.

Under a traditional case-cohort design where $q = 1$ and failure rate and dropout before the end of study are low, i.e., $n_k \approx n$, we note

$$\begin{aligned}
\sum_{k=1}^D \left\{ 1 - \sum_{k' \leq k} \frac{1}{n_{k'}} \right\}^2 &\approx \sum_{k=1}^D \left\{ 1 - \frac{k}{n} \right\}^2 \\
&= D - \frac{D(D+1)}{n} + \frac{D(D+1)(2D+1)}{6n^2}.
\end{aligned}$$

Thus, χ given in (2) is approximated by

$$p_D + \frac{1-p}{p} p_D^2 + o(p_D^2).$$

Then the power (1) can be approximated by

$$\Phi \left(Z_\alpha + \sqrt{np\theta} \sqrt{\frac{\beta\gamma(1-\gamma)p_D}{p + (1-p)p_D}} \right).$$

This reduces to the formula given in Cai and Zeng (2004).

4. Numerical Studies

We conduct simulations to examine the small sample performance of the proposed test. We vary the choices of p and q in the simulations. The results will be compared to the full-cohort design to study the efficiency loss due to the design. Especially, we study different scenarios including different failure rates and censoring scenarios. For each study, we report

the lower and the upper bounds of W_n 's power based on the calculation given in the previous section. These bounds are compared with the empirical power from 5000 samples and with the power in the full-cohort design. For secondary interest, the powers based on the formula in Cai and Zeng (2004) and from the full cohort are also given.

In all the studies, we assume that two groups have equal sizes ($\gamma = 1/2$) and that subjects in both groups have the same censoring pattern, thus $\beta = 1$. Furthermore, censoring time is from a mixture distribution with probability p_c being generated from uniform distribution in $[0, \tau]$ and probability $(1 - p_c)$ being τ . Then by choosing $p_c = 0$, we obtain the situation where no premature dropout occurs; while by choosing $p_c > 0$, we obtain the situation where premature dropout exists.

Tables 1 and 2 report the simulation results from $n = 1000$ and 2000, respectively. For different failure rates of 20% and 40%, each table lists the results from different sampling fractions and different effects under the situation with or without premature drop outs. The two numbers in the parentheses are the lower bounds and the upper bounds using our proposed calculation. Column "Full" is the empirical power of log-rank statistic using the full-cohort design; column "Emp. Power" gives the empirical powers based on the test statistic W_n from the simulations; column "Rare" gives the power calculated using the rare-disease formula given in Cai and Zeng (2004).

It is observed from the two tables that the intervals based on our calculation are fairly precise and close to the empirical values. The lengths of the intervals tend to be larger with increase of failure rates but all are smaller than 0.1. When the failure rate is high, we can sample only half of the additional cases, i.e., $q = 0.5$, and achieve similar power as the traditional case-cohort design where $q = 1$. Interestingly, we also observe that the calculated power using the formula in Cai and Zeng (2004), even though it is for rare event, has a good approximation to the empirical value when the event is in fact not rare.

Finally, we illustrate our power calculation via the Atherosclerosis Risk in Communities (ARIC) study (ARIC Investigator, 1989). The study contains a biethnic cohort of men and women under active follow-up since 1987–1989. Four endpoints quantifying subclinical atherosclerosis and validated clinical atherosclerotic events are being studied under the case-cohort design. We particularly focus on all incident coronary heart disease cases occurring between 1987 and 2001 among whites who have ever smoked. The total size of this subpopulation was 6901, of which 872 had coronary heart disease cases, so $p_D = 0.1264$. A subcohort of 921 ($p = 0.1335$) subjects were selected. In this illustration, we specially focus on the polymorphism on one specific single-nucleotide polymorphism of a major repair gene, whose frequencies of genotype (AA) versus the other genotypes were assumed to be 0.8 versus 0.2, so $\gamma = 0.8$. Assuming that the risk effect is 0.4 ($\theta = 0.4$), and the two groups have the same censoring pattern ($\beta = 1$), we can use formula (1) to calculate powers under different failure sampling fractions, where the risk sizes n_k 's in (1) at each failure can be obtained from the data. Figure 1 plots this trend and it indicates that with only 30% failure sampling fraction ($q = 0.3$) among the cases

Table 1
Simulation results from 5000 repetitions with $n = 1000$

$P(T < \tau)$	e^θ	p	p_c	Full	$q = 1$		$q = 0.5$		Rare ($q = 1$)
					Emp. power	Bounds	Emp. power	Bounds	
20%	1	0.1	0	0.051	0.050	(0.050, 0.050)	0.052	(0.050, 0.050)	0.050
			0.2	0.054	0.050	(0.050, 0.050)	0.052	(0.050, 0.050)	0.050
		0.2	0	0.051	0.052	(0.050, 0.050)	0.058	(0.050, 0.050)	0.050
			0.2	0.052	0.059	(0.050, 0.050)	0.063	(0.050, 0.050)	0.050
	1.25	0.1	0	0.301	0.200	(0.197, 0.200)	0.165	(0.163, 0.165)	0.199
			0.2	0.277	0.188	(0.177, 0.190)	0.152	(0.150, 0.156)	0.190
		0.2	0	0.301	0.249	(0.235, 0.238)	0.193	(0.185, 0.186)	0.237
			0.2	0.277	0.232	(0.212, 0.222)	0.181	(0.170, 0.174)	0.222
	1.50	0.1	0	0.610	0.419	(0.407, 0.415)	0.327	(0.318, 0.322)	0.413
			0.2	0.554	0.377	(0.358, 0.390)	0.297	(0.285, 0.299)	0.391
		0.2	0	0.610	0.510	(0.494, 0.500)	0.389	(0.370, 0.374)	0.498
			0.2	0.555	0.458	(0.441, 0.464)	0.353	(0.335, 0.344)	0.464
40%	1	0.1	0	0.047	0.051	(0.050, 0.050)	0.052	(0.050, 0.050)	0.050
			0.2	0.050	0.050	(0.050, 0.050)	0.050	(0.050, 0.050)	0.050
		0.2	0	0.047	0.048	(0.050, 0.050)	0.047	(0.050, 0.050)	0.050
			0.2	0.050	0.050	(0.050, 0.050)	0.048	(0.050, 0.050)	0.050
	1.25	0.1	0	0.489	0.242	(0.229, 0.243)	0.201	(0.204, 0.212)	0.243
			0.2	0.440	0.228	(0.205, 0.234)	0.198	(0.185, 0.203)	0.235
		0.2	0	0.489	0.330	(0.308, 0.322)	0.268	(0.259, 0.266)	0.322
			0.2	0.440	0.304	(0.275, 0.305)	0.252	(0.234, 0.249)	0.306
	1.50	0.1	0	0.874	0.522	(0.493, 0.521)	0.445	(0.428, 0.446)	0.521
			0.2	0.830	0.482	(0.434, 0.503)	0.407	(0.379, 0.422)	0.503
		0.2	0	0.874	0.674	(0.656, 0.679)	0.565	(0.551, 0.563)	0.679
			0.2	0.830	0.627	(0.590, 0.647)	0.527	(0.496, 0.526)	0.648

Table 2
Simulation results from 5000 repetitions with $n = 2000$

$P(T < \tau)$	e^θ	p	p_c	Full	$q = 1$		$q = 0.5$		Rare ($q = 1$)
					Emp. power	Bounds	Emp. power	Bounds	
20%	1	0.1	0	0.047	0.047	(0.050, 0.050)	0.046	(0.050, 0.050)	0.050
			0.2	0.047	0.049	(0.050, 0.050)	0.050	(0.050, 0.050)	0.050
		0.2	0	0.047	0.049	(0.050, 0.050)	0.050	(0.050, 0.050)	0.050
			0.2	0.047	0.049	(0.050, 0.050)	0.051	(0.050, 0.050)	0.050
	1.25	0.1	0	0.458	0.301	(0.298, 0.304)	0.244	(0.239, 0.241)	0.303
			0.2	0.415	0.282	(0.263, 0.286)	0.225	(0.215, 0.225)	0.288
		0.2	0	0.458	0.365	(0.365, 0.370)	0.289	(0.278, 0.280)	0.369
			0.2	0.415	0.337	(0.325, 0.342)	0.264	(0.252, 0.258)	0.344
	1.50	0.1	0	0.854	0.635	(0.632, 0.643)	0.518	(0.503, 0.509)	0.643
			0.2	0.797	0.583	(0.570, 0.617)	0.465	(0.453, 0.477)	0.613
		0.2	0	0.854	0.744	(0.743, 0.749)	0.584	(0.585, 0.588)	0.749
			0.2	0.797	0.689	(0.684, 0.712)	0.541	(0.532, 0.545)	0.710
40%	1	0.1	0	0.044	0.050	(0.050, 0.050)	0.047	(0.050, 0.050)	0.050
			0.2	0.043	0.050	(0.050, 0.050)	0.048	(0.050, 0.050)	0.050
		0.2	0	0.044	0.053	(0.050, 0.050)	0.051	(0.050, 0.050)	0.050
			0.2	0.043	0.051	(0.050, 0.050)	0.046	(0.050, 0.050)	0.050
	1.25	0.1	0	0.729	0.379	(0.357, 0.380)	0.333	(0.313, 0.327)	0.380
			0.2	0.680	0.354	(0.314, 0.367)	0.306	(0.278, 0.311)	0.367
		0.2	0	0.729	0.514	(0.490, 0.512)	0.427	(0.408, 0.419)	0.512
			0.2	0.680	0.472	(0.435, 0.485)	0.380	(0.366, 0.390)	0.486
	1.50	0.1	0	0.987	0.762	(0.745, 0.776)	0.685	(0.665, 0.688)	0.776
			0.2	0.976	0.718	(0.673, 0.757)	0.638	(0.598, 0.658)	0.756
		0.2	0	0.987	0.902	(0.895, 0.910)	0.817	(0.806, 0.818)	0.909
			0.2	0.976	0.871	(0.843, 0.889)	0.771	(0.748, 0.781)	0.888

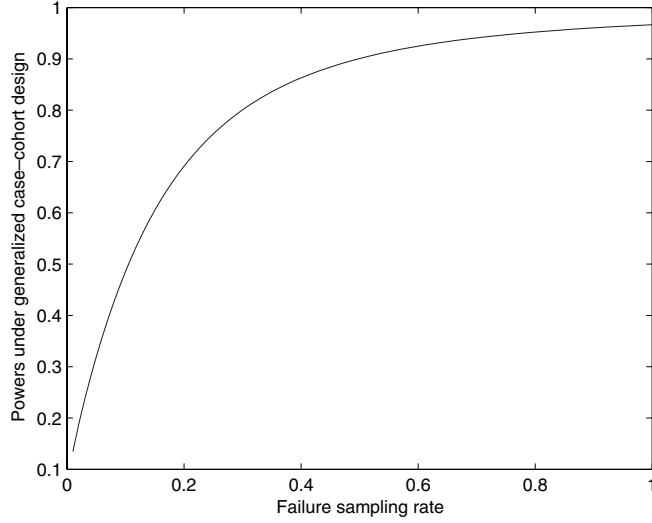


Figure 1. Plot of powers under the generalized case-cohort design versus failure sampling fractions in the ARIC example.

outside the subcohort, we can have 80% power to detect the difference between two groups.

5. Discussion

The case-cohort design is traditionally used for rare events in epidemiologic studies. However, with large number of subjects in studies and budget limitation of collecting risk-factor information from subjects, as seen in recent gene-disease association studies, case-cohort design can be applied to non-rare events as well. In this article, we propose a generalized case-cohort design, where instead of all the failures, only a fraction of these subjects are chosen for collection of additional exposure assessment. Our test statistic for this generalized design is shown to be valid numerically and the efficiency loss compared to the traditional case-cohort design is very low. We have also proposed a useful way to compute the power for such design.

An alternative design (DesignA1) is to sample a fraction p for the subcohort and then sample a fraction q of all the cases from the entire cohort. This design on average generates the same number of cases and noncases as the proposed design. Using the notation of ξ and η , the test statistic is the same as the proposed test statistic. However, following the proof, the variance of the test statistic for DesignA1 is given by

$$\begin{aligned} \tilde{\sigma}_W^2 = & \sigma^2 + \frac{1-p}{p} \text{Var}(\nu(Z, X)(1-\Delta)) \\ & + \frac{(1-q)}{q} \text{Var}(\mu(Z, X)\Delta(1-\xi)), \end{aligned}$$

which is greater than σ_W^2 in Theorem 1 by noting that

$$\begin{aligned} \text{Var}(\mu(Z, X)\Delta(1-\xi)) & \geq E[\text{Var}(\mu(Z, X) | \Delta, \xi)] \\ & \geq P(\Delta = 1)(1-p) \text{Var}(\mu(Z, X) | \Delta, \xi). \end{aligned}$$

The main reason is that η , the indicator of sampling cases, is from the full cohort but such an indicator in the proposed design is from the subjects outside the subcohort; intuitively,

conditioning on sampling out the subcohort, the number of cases outside of the subcohort is random in DesignA1 as compared to being fixed in the proposed design. Because of the smaller variance one would expect that the proposed design will give a larger power. Another alternative design (DesignA2), which was noted by a referee, is to sample a fraction of the subjects from the full cohort, and then obtain the standard case-cohort design using this subset. In this design, when the numbers of effective cases and noncases are on average the same as the proposed design, we can show that the asymptotic variance of the test statistic is equal to

$$\begin{aligned} \tilde{\sigma}_W^2 = & \sigma^2 + \frac{1-p}{p} \text{Var}(\nu(Z, X)(1-\Delta)) \\ & + \frac{(1-q)(1-p)}{p+q-pq} \text{Var}(\mu(Z, X)\Delta). \end{aligned}$$

This variance can be larger or smaller than the variance from the proposed design. We conducted some additional simulations using the same setup as in the previous simulations to compare these three designs. The results are summarized in Table 3. The results show that the proposed design has slightly larger power than DesignA1 but can be slightly larger or smaller than DesignA2, which confirms the above theoretical derivation. The similar power of the three designs also suggests that one could use the sample size/power formulae developed for the proposed design as an approximation for the other two alternative designs.

In practice, a budget is often restricted or the stored raw material is limited for collection of expensive risk-factor information. Suppose that due to budget restrictions, at most N_0 subjects can be selected for this assessment. Then we obtain $pn + (1-p)qD \leq N_0$. One interesting question is how to choose optimal (p, q) , the sampling fractions, to obtain the largest power. We recommend that the optimal (p, q) should be the one minimizing $\chi_U(p, q)$, where $\chi_U(p, q)$ is the upper bound for (2) as defined in Section 3. Note that $\chi_U(p, q)$ is decreasing in q . The following procedure can be used to find the optimal (p, q) : let p take value from a grid partition of $[0, 1]$ such that $pn \leq N_0$; for each p , let $q = \min\{1, (N_0 - pn)/((1-p)D)\}$ then calculate $\chi_U(p, q)$. The optimal p_{opt} should be the grid point with the smallest $\chi_U(p, q)$ and the optimal q_{opt} is given by $\min\{1, (N_0 - p_{\text{opt}}n)/((1-p_{\text{opt}})D)\}$. We can illustrate this calculation using the numbers in the real data example in Section 4. In that example, the subcohort size was 921. Based on the results in that example, if we sample 30% of the cases outside the subcohort, we have the total case-cohort sample size to be 1148. Figure 2 shows how the powers vary with different choices of p if we hold $N_0 = 1148$. It turns out that the optimal design should be $p_{\text{opt}} = 0.09$ and $q_{\text{opt}} = 0.66$. Thus, our calculation shows that under the same financial constraint, i.e., if we hold the total case-cohort sample size the same, the power would have been larger if we had reduced the subcohort size but increased the sampled cases outside the subcohort.

In some case-cohort studies, the expensive covariate could be a continuous variable. To utilize our formulae, one could choose a threshold value to categorize the covariate into two groups, and then examine the power based on our proposed statistic. However, in some studies, the original variable needs

Table 3
Comparison of empirical powers from 5000 repetitions

$P(T < \tau)$	e^θ	p	p_c	$n = 1000$			$n = 2000$		
				Proposed design	DesignA1	DesignA2	Proposed design	DesignA1	DesignA2
20%	1	0.1	0	0.052	0.050	0.054	0.046	0.045	0.045
			0.2	0.052	0.049	0.054	0.050	0.046	0.046
		0.2	0	0.058	0.053	0.048	0.050	0.048	0.043
			0.2	0.063	0.058	0.048	0.051	0.046	0.043
	1.25	0.1	0	0.165	0.163	0.173	0.244	0.243	0.241
			0.2	0.152	0.148	0.162	0.225	0.225	0.219
		0.2	0	0.193	0.187	0.187	0.289	0.285	0.273
			0.2	0.181	0.179	0.170	0.264	0.259	0.250
	1.50	0.1	0	0.327	0.326	0.340	0.518	0.503	0.498
			0.2	0.297	0.290	0.311	0.465	0.461	0.465
		0.2	0	0.389	0.384	0.383	0.584	0.581	0.588
			0.2	0.353	0.343	0.344	0.541	0.535	0.546
40%	1	0.1	0	0.052	0.052	0.053	0.047	0.048	0.044
			0.2	0.050	0.050	0.053	0.048	0.049	0.043
		0.2	0	0.047	0.050	0.045	0.051	0.050	0.045
			0.2	0.048	0.048	0.042	0.046	0.049	0.046
	1.25	0.1	0	0.201	0.207	0.218	0.333	0.332	0.318
			0.2	0.198	0.194	0.207	0.306	0.306	0.289
		0.2	0	0.268	0.267	0.258	0.427	0.422	0.408
			0.2	0.252	0.250	0.231	0.380	0.380	0.379
	1.50	0.1	0	0.445	0.439	0.462	0.685	0.679	0.674
			0.2	0.407	0.407	0.422	0.638	0.637	0.636
		0.2	0	0.565	0.564	0.552	0.817	0.811	0.810
			0.2	0.527	0.517	0.508	0.771	0.766	0.765

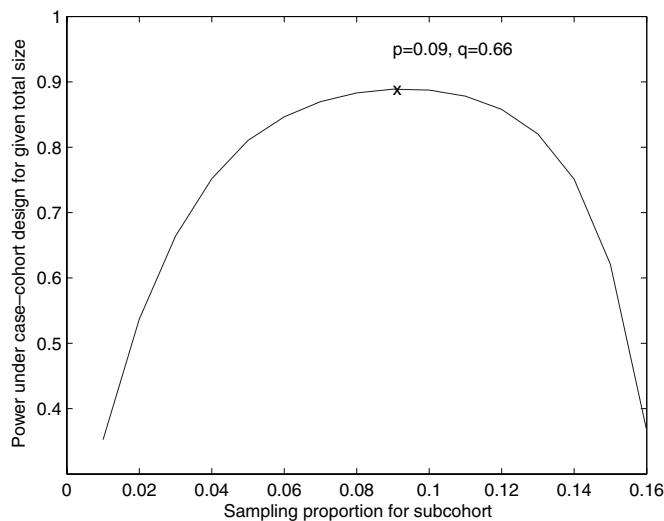


Figure 2. Plot of powers under the generalized case-cohort design versus sampling proportions for subcohort when total number of subjects for genotype assessment is 1148 in the ARIC study.

to be kept. Under this situation, the power calculation is much more complicated. For the traditional case-cohort design ($q = 1$), one possible approach is to base on a pseudoscore test statistic as given in Wacholder, Gail, and Pee (1991): let

n_k denote the risk set size at time t_k and let σ_x^2 be the variance of covariate. Wacholder, et al. (1991) obtained the noncentral parameter of the pseudoscore test given by

$$\theta / \sqrt{\left\{ \sum_{k=1}^D V_{kk} + 2 \sum_{k < j, k, j=1}^D V_{kj} \right\} / \left\{ \sum_{k=1}^D V_{kk} \right\}^2},$$

where V_{kk} is approximated by $\sigma_x^2 \{(n_k - 1)p\} / \{(n_k - 1)p + 1\}$ and for $k < j$, V_{kj} is approximated by $\sigma_x^2 (p - p^2)(n_j - 1) / \{(n_k - 1)p + 1\} \{(n_j - 1)p + 1\}$. We can further simplify this expression assuming that the number of subjects at the end of study is much larger than 1. Because $V_{kk} \approx \sigma_x^2$ and $V_{kj} \approx \sigma_x^2 (1 - p) / \{p(n_k - 1)\}$, simple algebra gives that the power should be equal to

$$\tilde{P} = \Phi \left(Z_\alpha + \frac{\theta \sigma_x D / \sqrt{n}}{\sqrt{p_D + \frac{1-p}{p} \frac{1}{n} \sum_{k=1}^D \frac{2D-2k}{n_k}}} \right).$$

Thus, when X is continuous with variance σ_x^2 , as before, from the fact that n_k is in between $(n - n_c - D)$ and $(n - k + 1)$, we obtain the lower bound and upper bound for the power which is given as

$$\Phi \left(Z_\alpha + \frac{\theta \sigma_x D / \sqrt{n}}{\sqrt{p_D + \frac{1-p}{np} \sum_{k=1}^D \frac{2D-2k}{n-D-n_c}}} \right) \leq \tilde{p} \leq \Phi \left(Z_\alpha + \frac{\theta \sigma_x D / \sqrt{n}}{\sqrt{p_D + \frac{1-p}{np} \sum_{k=1}^D \frac{2D-2k}{n-k+1}}} \right). \quad (3)$$

Additionally, when disease is rare and drop out is rare, both lower bound and upper bound are equal to

$$\Phi \left(Z_\alpha + \sqrt{np} \theta \sqrt{\frac{\sigma_x^2 p_D}{p + (1-p)p_D}} \right). \quad (4)$$

It will be interesting to examine whether (3) and (4) also apply to our generalized design. For estimating the effects of covariates in this generalized design, one possible method is to maximize a generalized version of the pseudopartial likelihood function in Prentice (1986)

$$\prod_{i=1}^n \left[\frac{\exp \{X_i^T \beta\}}{\sum_{j=1}^n (\xi_j \Delta_j + \xi_j (1 - \Delta_j) / p + (1 - \xi_j) \Delta_j \eta_j / q) I(Y_j \geq Y_i) \exp \{X_j^T \beta\}} \right]^{\Delta_i (\xi_i + (1 - \xi_i) \eta_i / q)},$$

where the risk set in the whole cohort is approximated by the sampled subjects. The asymptotic variance can be estimated using the sandwich-type estimate. We will pursue this approach in future work.

6. Supplementary Materials

The proof of Theorem 1 and the technical simplification of σ_W^2 in Section 3 are given in the appendices available under the Paper Information link at the Biometrics website: <http://www.biometrics.tibs.org>.

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A.1 Proof of Theorem 1

To derive the asymptotic distribution of $W_n(t)$, we define $Y_j(t) = \sum_{i=1}^n I(X_i = j, Z_i \geq t)$ and $N_j(t) = \sum_{i=1}^n I(X_i = j, Z_i \leq t)$, i.e., the risk set and the counting process of group j in the full cohort. Thus, using the similar expansion as given in Cai and Zeng (2004) and the fact that $\tilde{Y}_j(t) = Y_j(t) + O_p(n^{1/2})$ and $\tilde{N}_j(t) = N_j(t) + O_p(n^{1/2})$ uniformly in $t \in [0, \tau]$, we obtain

$$W_n = \int_0^\tau \frac{\omega(t)Y_1(t)Y_2(t)}{Y_1(t) + Y_2(t)} \left[\frac{dN_1(t)}{Y_1(t)} - \frac{dN_2(t)}{Y_2(t)} \right] + \sum_{i=1}^n \mu(Z_i, X_i) \Delta_i (1 - \xi_i) \{ \eta_i - q \} / q + \sum_{i=1}^n \nu(Z_i, X_i) (1 - \Delta_i) \{ \xi_i - p \} / p + o_p(n^{1/2}).$$

Clearly, the above expression consists three parts: the first part is the usual log-rank statistic in the full cohort; the second part is due the random sampling of the subcohort in the design; the third part is due to the random sampling of additional cases in the design. The Hajek’s theorem (c.f. Self and Prentice, 1988) for sampling without replacement can be applied to show that these parts are asymptotically independent and moreover,

$$\frac{1}{\sqrt{\tilde{n}}} \sum_{i=1}^n \mu(Z_i, X_i) \Delta_i (1 - \xi_i) (\eta_i - q) / q \rightarrow_d N(0, (1 - q)q^{-1} \text{Var}(\mu(Z, X) | \Delta = 1, \xi = 0)),$$

and

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \nu(Z_i, X_i) (1 - \Delta_i) \{ \xi_i - p \} / p \rightarrow_d N(0, (1 - p)p^{-1} \text{Var}(\nu(Z, X) (1 - \Delta))).$$

Consequently, from the asymptotic property of the usual log-rank statistic, we conclude that $n^{-1/2}W_n$ has an asymptotically normal distribution with asymptotic bias under the alternative the same as the usual log-rank test while its variance equal to

$$\sigma^2 + \frac{1 - p}{p} \text{Var}(\nu(Z, X) (1 - \Delta)) + \frac{P(\Delta = 1)(1 - p)(1 - q)}{q} \text{Var}(\mu(Z, X) | \Delta = 1, \xi = 0),$$

where σ^2 is the asymptotic variance of the usual log-rank test statistic.

A.2 Simplification of σ_W^2

From the explicit expressions of $\nu(Z, X)$, we can obtain

$$\begin{aligned} & Var(\nu(Z, X)(1 - \Delta)) \\ &= \gamma \int_0^\tau e^{-\Lambda(t)} \left\{ \int_0^t (1 - \alpha(s)) d\Lambda(s) \right\}^2 dP(C \leq t | X = 1) \\ &\quad + (1 - \gamma) \int_0^\tau e^{-\Lambda(t)} \left\{ \int_0^t \alpha(s) d\Lambda(s) \right\}^2 dP(C \leq t | X = 2) \\ &\quad - \left[-\gamma \int_0^\tau e^{-\Lambda(t)} \int_0^t (1 - \alpha(s)) d\Lambda(s) dP(C \leq t | X = 1) \right. \\ &\quad \left. + (1 - \gamma) \int_0^\tau e^{-\Lambda(t)} \int_0^t \alpha(s) d\Lambda(s) dP(C \leq t | X = 2) \right]^2. \end{aligned}$$

After integration by part, we have

$$\begin{aligned} & Var(\nu(Z, X)(1 - \Delta)) \\ &= \gamma \int_0^\tau P(C > t | X = 1) \left\{ -\left(\int_0^t (1 - \alpha(s)) d\Lambda(s) \right)^2 + 2 \int_0^t (1 - \alpha(s)) d\Lambda(s) (1 - \alpha(t)) \right\} e^{-\Lambda(t)} d\Lambda(t) \\ &\quad + (1 - \gamma) \int_0^\tau P(C > t | X = 2) \left\{ -\left(\int_0^t \alpha(s) d\Lambda(s) \right)^2 + 2 \int_0^t \alpha(s) d\Lambda(s) \alpha(t) \right\} e^{-\Lambda(t)} d\Lambda(t) \\ &\quad - \left[-\gamma \int_0^\tau P(C > t | X = 1) \left\{ -\int_0^t (1 - \alpha(s)) d\Lambda(s) + (1 - \alpha(t)) \right\} e^{-\Lambda(t)} d\Lambda(t) \right. \\ &\quad \left. + (1 - \gamma) \int_0^\tau P(C > t | X = 2) \left\{ -\int_0^t \alpha(s) d\Lambda(s) + \alpha(t) \right\} e^{-\Lambda(t)} d\Lambda(t) \right]^2. \end{aligned}$$

Similarly,

$$\begin{aligned} & Var(\mu(Z, X) | \Delta = 1, \xi = 0) \\ &= \frac{1}{P(\Delta = 1)} \left[\gamma \int_0^\tau P(C > t | X = 1) \left\{ (1 - \alpha(t)) - \int_0^t (1 - \alpha(s)) d\Lambda(s) \right\}^2 e^{-\Lambda(t)} d\Lambda(t) \right. \\ &\quad \left. + (1 - \gamma) \int_0^\tau P(C > t | X = 2) \left\{ -\alpha(t) + \int_0^t \alpha(s) d\Lambda(s) \right\}^2 e^{-\Lambda(t)} d\Lambda(t) \right] \\ &\quad - \frac{1}{P(\Delta = 1)^2} \left[\gamma \int_0^\tau P(C > t | X = 1) \left\{ (1 - \alpha(t)) - \int_0^t (1 - \alpha(s)) d\Lambda(s) \right\} e^{-\Lambda(t)} d\Lambda(t) \right. \\ &\quad \left. + (1 - \gamma) \int_0^\tau P(C > t | X = 2) \left\{ -\alpha(t) + \int_0^t \alpha(s) d\Lambda(s) \right\} e^{-\Lambda(t)} d\Lambda(t) \right]^2. \end{aligned}$$

Moreover, since

$$\sigma^2 = \int_0^\tau \left\{ \alpha(t)^2(1-\gamma)P(C > t|X=2) + (1-\alpha(t))^2\gamma P(C > t|X=1) \right\} e^{-\Lambda(t)} d\Lambda(t),$$

we conclude that the asymptotic variance of $n^{-1/2}W_n$ is equal to

$$\begin{aligned} \sigma_W^2 &= \int_0^\tau P(C > t|X=1) \left[\gamma \left(1 + \frac{(1-p)(1-q)}{q} \right) (1-\alpha(t))^2 \right. \\ &\quad + 2\gamma \left(\frac{1-p}{p} - \frac{(1-p)(1-q)}{q} \right) \int_0^t (1-\alpha(s)) d\Lambda(s) (1-\alpha(t)) \\ &\quad \left. + \gamma \left(-\frac{1-p}{p} + \frac{(1-p)(1-q)}{q} \right) \left\{ \int_0^t (1-\alpha(s)) d\Lambda(s) \right\}^2 \right] e^{-\Lambda(t)} d\Lambda(t) \\ &\quad + \int_0^\tau P(C > t|X=2) \left[(1-\gamma) \left(1 + \frac{(1-p)(1-q)}{q} \right) \alpha(t)^2 \right. \\ &\quad + 2(1-\gamma) \left(\frac{1-p}{p} - \frac{(1-p)(1-q)}{q} \right) \int_0^t \alpha(s) d\Lambda(s) \alpha(t) \\ &\quad \left. + (1-\gamma) \left(-\frac{1-p}{p} + \frac{(1-p)(1-q)}{q} \right) \left\{ \int_0^t \alpha(s) d\Lambda(s) \right\}^2 \right] e^{-\Lambda(t)} d\Lambda(t) \\ &\quad - \left\{ \frac{1-p}{p} + \frac{(1-p)(1-q)}{P(\Delta=1)q} \right\} \\ &\quad \times \left[\gamma \int_0^\tau P(C > t|X=1) \left\{ (1-\alpha(t)) - \int_0^t (1-\alpha(s)) d\Lambda(s) \right\} e^{-\Lambda(t)} d\Lambda(t) \right. \\ &\quad \left. + (1-\gamma) \int_0^\tau P(C > t|X=2) \left\{ -\alpha(t) + \int_0^t \alpha(s) d\Lambda(s) \right\} e^{-\Lambda(t)} d\Lambda(t) \right]^2, \end{aligned}$$

where we also note that

$$P(\Delta=1) = \gamma \int_0^\tau P(C > t|X=1) e^{-\Lambda(t)} d\Lambda(t) + (1-\gamma) \int_0^\tau P(C > t|X=2) e^{-\Lambda(t)} d\Lambda(t).$$

Under the assumption $P(C > t|X=2) = \beta P(C > t|X=1)$, $\alpha(t) = \gamma / \{\gamma + (1-\gamma)\beta\}$.

Then the last term in the above expression of σ_W^2 is zero and σ_W^2 can be simplified as

$$\begin{aligned} \sigma_W^2 &= \frac{\beta\gamma(1-\gamma)}{\gamma + (1-\gamma)\beta} \left\{ 1 + \frac{(1-p)(1-q)}{q} \right\} \int_0^\tau S_c(t) e^{-\Lambda(t)} d\Lambda(t) \\ &\quad + 2 \frac{\beta\gamma(1-\gamma)}{\gamma + (1-\gamma)\beta} \left\{ \frac{1-p}{p} - \frac{(1-p)(1-q)}{q} \right\} \int_0^\tau S_c(t) \Lambda(t) e^{-\Lambda(t)} d\Lambda(t) \\ &\quad + \frac{\beta\gamma(1-\gamma)}{\gamma + (1-\gamma)\beta} \left\{ -\frac{1-p}{p} + \frac{(1-p)(1-q)}{q} \right\} \int_0^\tau S_c(t) \Lambda(t)^2 e^{-\Lambda(t)} d\Lambda(t). \end{aligned}$$